

**DEPARTMENT OF CHEMICAL & BIOMOLECULAR ENGINEERING  
NORTH CAROLINA STATE UNIVERSITY**

**CHE 330      EVALUATION OF A DOUBLE-PIPE HEAT EXCHANGER<sup>†</sup>**

**1. Introduction**

Heat transfer is a common, important unit operation in the process industry. There are three modes of heat transfer: conduction, convection, and radiation. Conduction is the transfer of heat across a solid or stationary layer of a material. Convection occurs when heat is transferred via the motion of a fluid, either under a temperature-induced density gradient (free convection) or a pressure gradient (forced convection). Radiation is heat transfer via electromagnetic waves. Although radiation often involves a high-temperature body (e.g. the sun), high temperature is not a necessary attribute for radiation; for instance, the nightly heat loss from home occurs primarily through infrared radiation.

Efficient design of a heat transfer process requires a good understanding of the governing parameters. For convective and conductive heat transfer, the heat transfer rate  $Q$  is governed by the heat transfer area ( $A$ ), the temperature driving force ( $\Delta T$ ), and the overall heat transfer coefficient ( $U$ ) that is a composite of the thermal conductivity ( $k$ ) of the material and local convective heat transfer coefficient ( $h$ ) at one or more phase boundary (see **Theory**). For a specified  $Q$ , a lower equipment cost—which corresponds to a small heat transfer area  $A$ —is associated with either a high  $\Delta T$  or a high  $U$ .

Though not much used industrially, the double-pipe heat exchanger provides a simple, convenient and clear-cut model for studying the key factors affecting convective and conductive heat transfer.

**2. Objective**

The objective of this experiment is to determine the heat transfer and resistance characteristics of a single-pass, double-pipe heat exchanger that can operate in a co-current or counter-current mode.

**3. Theory**

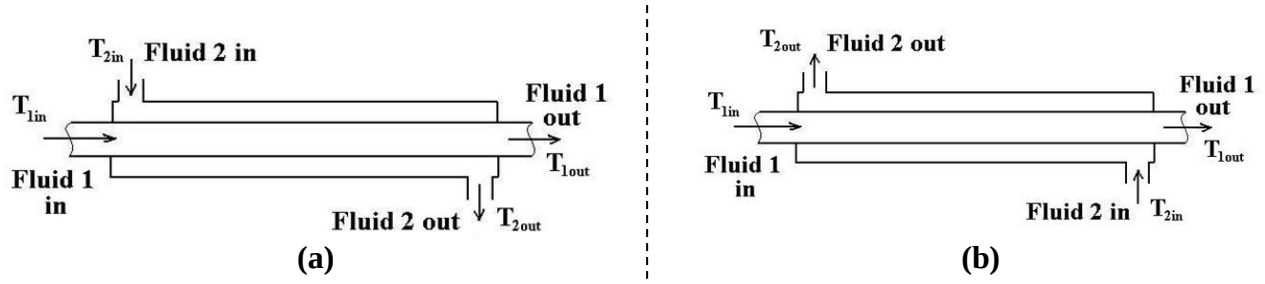
A simple schematic of a double-pipe heat exchanger is given in Figure 1. In the studied case, hot fluid flows through the inner tube of the heat exchanger, while cold water flows in the outer tube (i.e. the “shell”). Heat is exchanged between the streams through the wall of the inner pipe. For a double-pipe heat exchanger operating at steady state, the rate of heat transfer ( $Q$ ) may be described by the following general heat transfer equations [1]:

$$\dot{Q} = -\dot{m}\hat{C}_p(T_e - T_o) = \dot{m}'\hat{C}_p'(T_e' - T_o') = UA\Delta T_{\ln} \quad (1)$$

where  $\dot{m}$  and  $\dot{m}'$  are the mass flow rates of the hot and cold streams,  $\hat{C}_p$  and  $\hat{C}_p'$  are the specific

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**Figure 1. Schematic of double-pipe heat exchanger operation in (a) co-current and (b) counter-current heat transfer configurations [2].**

heat capacities of the hot and cold streams, and  $\Delta T_{\ln}$  is the log mean temperature difference:

$$\Delta T_{\ln} \equiv \text{log mean temperature difference} \equiv \frac{\Delta T_{\text{one end}} - \Delta T_{\text{other end}}}{\ln \frac{\Delta T_{\text{one end}}}{\Delta T_{\text{other end}}}} \quad (2)$$

Depending on whether the hot and cold water flow in the same direction (co-current heat exchange configuration) as shown in Figure 1 (a), or opposite directions (counter-current heat exchange configuration) as shown in Figure 1 (b),  $\Delta T_{\ln}$  is defined differently:

- **Co-current Heat Exchange:**

$$\Delta T_{\ln} = \frac{(T_o - T_o') - (T_e - T_e')}{\ln \frac{(T_o - T_o')}{(T_e - T_e')}} \quad (3)$$

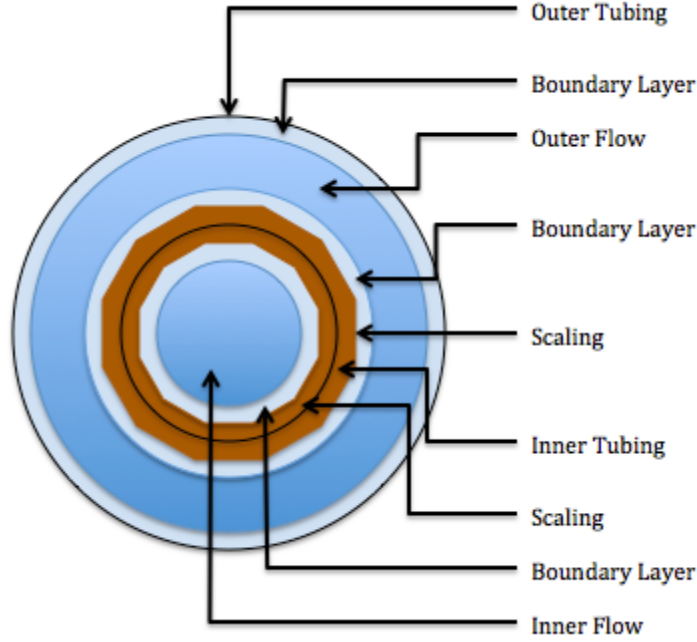
- **Counter-current Heat Exchange:**

$$\Delta T_{\ln} = \frac{(T_o - T_e') - (T_e - T_o')}{\ln \frac{(T_o - T_e')}{(T_e - T_o')}} \quad (4)$$

where  $T_o$  and  $T_o'$  are the inlet temperatures of the hot and cold streams, while  $T_e$  and  $T_e'$  are the outlet temperatures of the hot and cold streams [1].

### 3.1. Wilson Method of Analysis

A key driver in heat exchange design is the overall heat transfer coefficient ( $U$ ). Larger  $U$  values require less heat exchange area to transfer a given amount of heat. Understanding the factors influencing  $U$  requires a closer look at the path taken by heat as it transfers from the hot stream to the cold stream. The cross-section of a typical double-pipe heat exchanger is described in Figure 2. Since the hot stream in the inner pipe is at a higher temperature than the cold stream in the outer pipe, the resulting temperature gradient causes heat to transfer from the hot stream to the cold



**Figure 2. Cross-section of a typical double-pipe heat exchanger [3].**

stream. However, in real-life practice the transferring heat encounters significant resistance from a number of heat transfer resistors, as described by Equation 5:

$$\frac{1}{UA} = \frac{1}{h_i A_i} + \frac{1}{h_o A_o} + \frac{t_w}{k_w A_w} + \left( \frac{R_i}{A_i} + \frac{R_o}{A_o} \right) \quad (5)$$

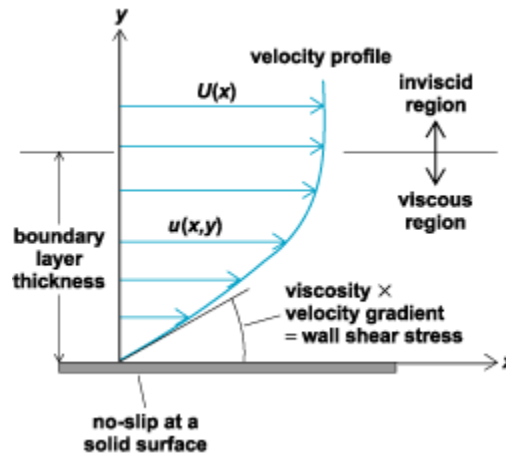
where  $h_i$  and  $h_o$  are the local convective heat transfer coefficients on the inside and outside surfaces of the heat exchange tube,  $A_i$  and  $A_o$  are the corresponding areas of the inside and outside heat exchange surfaces,  $R_i/A_i$  and  $R_o/A_o$  are the corresponding scale resistances on the inside and outside heat exchange surfaces,  $t_w$ ,  $k_w$  and  $A_w$  the wall thickness, thermal conductivity and mean surface area ( $= 0.5A_i + 0.5A_o$ ) of the material of construction of the heat exchange tube [1].

These heat transfer resistances (and Equation 5) are analogous to those for resistors in series in an electric circuit. With this in mind, the total heat transfer resistance between the hot and cold streams is equivalent to  $\frac{1}{UA}$ . Referencing Equation 5 and Figure 4, we can model the flow of energy as it travels from the hot stream to the cold stream, accounting for each heat transfer resistance that is overcome along the way:

- Convective heat transfer resistance in the hot stream  $\left( \frac{1}{h_i A_i} \right)$  – resistance encountered by heat as it transfers across the hot fluid's boundary layer (see proceeding discussion)
- Scale resistance on the hot side  $\left( \frac{R_i}{A_i} \right)$  – resistance to heat transfer due to build-up of minerals or other deposits on the inside of the inner tube wall

- Conductive resistance  $\left(\frac{t_w}{k_w A_w}\right)$  – heat transfer resistance associated with heat transfer through the metal pipe wall separating the hot and cold streams
- Scale resistance on the cold side  $\left(\frac{R_o}{A_o}\right)$  – resistance to heat transfer due to build-up of minerals or other deposits on the outside of the inner tube wall
- Convective heat transfer resistance in the cold stream  $\left(\frac{1}{h_o A_o}\right)$  – resistance encountered by heat as it transfers across the cold fluid’s boundary layer (see proceeding discussion)

Convective heat transfer resistance in a moving fluid arises from heat traveling across a transition region between the stagnant, stationary fluid next to the pipe wall and the free-flowing section of the stream unaffected by the wall (this region is described in transport phenomena as the “boundary layer”); this phenomenon is described in Figure 3. The size of the boundary layer is affected by the velocity of the fluid, the shape of the wall, and the physical properties of the fluid. In general, as fluid velocity goes up, boundary layer thickness goes down. Thick boundary layers provide more resistance to heat transfer than thin boundary layers. Boundary layer thickness is the primary determinant of local convective heat transfer coefficient for each stream in the heat exchanger [1].



**Figure 3. Model of fluid flow against a wall, showing boundary layer profile [4].**

On a theoretical basis, provided that the flow is in the fully-developed turbulent regime, the local convective heat transfer coefficient for a tubular surface may be estimated using the Dittus-Boelter correlation:

$$Nu = 0.023 Re^{0.8} Pr^{0.3} \quad (6)$$

where  $Nu$  is the Nusselt number,  $Re$  is the Reynolds number, and  $Pr$  is the Prandtl number. The Nusselt number gives the ratio of heat transfer by convection to heat transfer by conduction, and will therefore depend on the size of the boundary layers. Substitution of physical parameters for the well-known  $Nu$ ,  $Re$ , and  $Pr$  expressions gives the Dittus-Boelter correlation:

$$\frac{hD}{k} = 0.023 \left[ \frac{\rho v D}{\mu} \right]^{0.8} \left[ \frac{\widehat{C}_p \mu}{k} \right]^{0.3} \quad (7)$$

For flow in a tube,  $D$  in Equation 7 is given by  $D_i$ , the inside diameter of the tube (i.e. the outer diameter of the inner pipe minus twice the wall thickness). For flow in the annular space between concentric pipes, an estimate of the effective hydraulic diameter  $D_h$  can be made by treating the region as a tube with hydraulic area  $A_h$ :

$$A_h = \pi \left( \frac{OD_{outer} - 2t_{w,outer}}{2} \right)^2 - \pi \left( \frac{OD_{inner}}{2} \right)^2 \quad (8)$$

$$D_h = 2 \sqrt{\frac{A_h}{\pi}} \quad (9)$$

where  $OD_{outer}$  is the outer diameter of the outer pipe,  $t_{w,outer}$  is the wall thickness of the outer pipe, and  $OD_{inner}$  is the outer diameter of the inner pipe.

According to the correlation, if flow velocity ( $v$ ) is varied while all other variables are held constant,  $h$  should be proportional to  $v^{0.8}$ :

$$h = 0.023 \left( \frac{k^{0.7} \rho^{0.8} \widehat{C}_p^{0.3}}{\mu^{0.5} D^{0.2}} \right) v^{0.8} \equiv \alpha v^{0.8} \quad (10)$$

where the proportionality constant  $\alpha$  is dependent on the physical properties of the fluid and the temperature.

Equation 10 may be used in conjunction with Equation 7 to determine  $h_i$  or  $h_o$  from the experimental data. This is the basis of the Wilson method of analysis. If the flowrate inside the tube is varied while all other parameters are held constant, Equation 5 and 8 give

$$\frac{1}{UA} = \left[ \frac{1}{h_o A_o} + \frac{t_w}{k_w A_w} + \left( \frac{R_i}{A_i} + \frac{R_o}{A_o} \right) \right] + \frac{1}{\alpha_i A_i v_i^{0.8}} \equiv C_i + \frac{1}{\alpha_i A_i v_i^{0.8}} \quad (11)$$

It follows that a plot of  $\frac{1}{UA}$  versus  $\frac{1}{v_i^{0.8}}$  should yield a straight line with a slope equal to  $\frac{1}{\alpha_i A_i}$  and an intercept equal to  $C_i$ .

Similarly, if  $UA$  is measured as a function of the flowrate in the annulus space,

$$\frac{1}{UA} = \left[ \frac{1}{h_i A_i} + \frac{t_w}{k_w A_w} + \left( \frac{R_i}{A_i} + \frac{R_o}{A_o} \right) \right] + \frac{1}{\alpha_o A_o v_o^{0.8}} \equiv C_o + \frac{1}{\alpha_o A_o v_o^{0.8}} \quad (12)$$

A plot of  $\frac{1}{UA}$  versus  $\frac{1}{v_o^{0.8}}$  should yield a straight line with a slope equal to  $\frac{1}{\alpha_o A_o}$  and an intercept equal to  $C_o$ . The value of thermal resistance of the wall ( $t_w/(k_w A_w)$ ) can be calculated from the physical data. Once the values of  $\alpha_i A_i$ ,  $C_i$ ,  $\alpha_o A_o$ , and  $C_o$  are known, the value of total scale resistance ( $R_i/A_i + R_o/A_o$ ) may be determined. The values determined for  $\alpha_i$  and  $\alpha_o$  from the Wilson analysis may be compared with the corresponding values calculated from the physical properties of the fluid using Equation 10 [1].

#### 4. Equipment

The experimental setup is shown in Figures 4-7. The heat exchanger consists of two concentric stainless steel (Type 314) tubes with the dimensions specified in Table 1. The heat exchanger has the following provisions: (1) two sets of rotameters and mass flow meters to measure the flowrates of the heat-exchanging streams, (2) six thermistors to measure the temperatures of the streams at the inlets, outlets and mid-points of the heat exchanger, (3) a re-circulating hot water reservoir whose feed temperature is maintained constant by an electric heater in conjunction with a temperature controller, (4) a re-circulating chilled water reservoir whose feed temperature is maintained constant by the building's chiller, (5) a pump that circulates the hot stream through the heat exchanger and returns it to the reservoir tank, and (6) the heat exchange mode can be varied between countercurrent and cocurrent by varying the settings of the cold flow valves as shown in Figures 6 and 7.

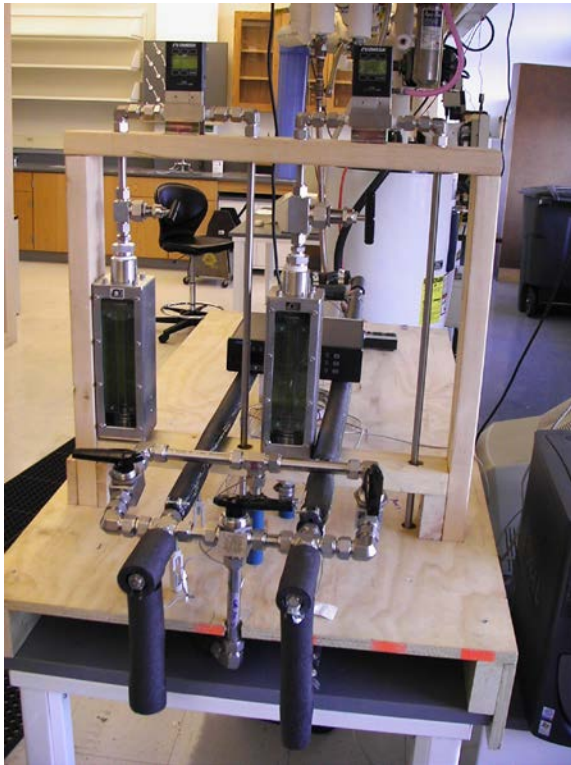


Figure 4. Heat exchanger experimental setup.



Figure 5. Thermistor readout panel.



Figure 6. Valve settings for counter-current flow. Figure 7. Valve settings for co-current flow.

Table 1. Dimensions of double-pipe heat exchanger pipes.

| Pipe  | Outside Diameter, m | Wall Thickness, m | Length, m | Heat Transfer Area, m <sup>2</sup> |                       |         |
|-------|---------------------|-------------------|-----------|------------------------------------|-----------------------|---------|
|       |                     |                   |           | Outer, A <sub>o</sub>              | Inner, A <sub>i</sub> | Average |
| Inner | 0.0127              | 0.00076           | 3.048     |                                    |                       |         |
| Outer | 0.0190              | 0.00124           | 3.048     | 0.1216                             | 0.1070                | 0.1143  |

Thermistor readings can be read off the thermistor panel by pressing the appropriate “Step” buttons successively. Thermistors 1, 2, and 3 monitor the temperatures of the hot stream at the inlet, outlet and mid-point, respectively, and they are activated by the left “Step” button on the panel. Thermistors 4, 5, and 6 monitor the temperatures of the cold stream at the inlet, outlet and mid-point, respectively, and they are activated by the upper right “Step” button on the panel. The “Step” buttons below the upper right “Step” button are not in service. For example, Figure 5 shows the output from Thermistor 1.

The calibration equation for the hot-flow and cold flow rotameters are given, respectively, by:

$$v_{hot-flow} = (1.296 * RR_{hot-flow} - 1.075) \frac{L}{min} \quad (13)$$

$$v_{cold-flow} = (1.029 * RR_{cold-flow} + 0.499) \frac{L}{min} \quad (14)$$

where RR denotes the rotameter readings.

## 5. Experimental Strategy and Procedure

The heat transfer characteristics of the heat exchanger are determined using the Wilson's method of analysis as described in the Theory section. Specifically, UA is measured as a function of the flowrates and inlet temperatures of the fluid in the inner tube and the annular space.

Observe safety precautions as noted in the Laboratory Safety Procedures. In particular, avoid contact with hot surfaces. Hot water is circulated through the inner tube of the heat exchanger by the pump and re-circulating cold water from a chilled water reservoir is fed, first cocurrently and then countercurrently, through the annular space between the concentric tubes. The flowrates of the streams are measured by means of the rotameters and flow meters and the inlet and outlet temperatures of the streams are measured by means of the thermistors. Care should be taken to establish that the flows are turbulent and that steady state has been reached before recording the data and going to a new run. How do you ensure that these two conditions are, in fact, met?

For each heat exchange mode, vary the hot and cold streams by 4 x 4 combinations for a total of 16 runs. Vary each of the hot and cold streams four times. Use the same set of four flowrates for the hot stream while the flowrate of the cold stream is varied four times (the hot and cold streams may, however, have different sets of four flowrates). Evenly space the four flowrate settings over as wide a flow range as possible to extend the range of the Wilson analysis. For each of the 16 combinations of settings, record the flowrates and the inlet and outlet temperatures of the two streams once steady state has been reached (typically 30-60 seconds). Then change the heat exchange mode and repeat the flowrate and temperature measurements for the same 4 x 4 variations of the hot and cold streams for another set of 16 runs. Prior to entering the lab to perform the experiment, prepare a spreadsheet/table containing the 16 flow combinations for each configuration (i.e., 16 flow combinations for the co-current configuration and 16 flow combinations for the counter-current flow configuration). Bring two copies of the table for use in the lab: one for the raw data collector, and one for the person changing the flow rates during the experiment.

To ensure that all 32 trials are completed in reasonable time, it is recommended that the heat exchange mode be kept the same until all 16 trials are complete and that the flow parameters are systematically varied one at a time. Avoid switching the heat exchange mode back and forth or varying more than one parameter at one time as these could introduce a large upset to the system and extend the wait time for the steady state. Whenever the heat exchange mode is switched, an extended wait time should be provided (about 15 minutes) for the first run to ensure that steady state is reached.

After changing the hot or cold water flowrate, a suggested time to allow the system to reach steady state is 30-60 seconds. From a theoretical perspective, at steady state  $Q_{hot}$  should equal  $Q_{cold}$ ; however, you may find that  $Q_{hot}$  and  $Q_{cold}$  are not quite equal in the laboratory. It can be reasonably assumed that steady state has been reached if  $Q_{hot}$  and  $Q_{cold}$  are within 15% of each other. As a confirmation that steady state has been reached, Equation 15 should be used:

$$\frac{|Q_{hot} - Q_{cold}|}{Q_{avg}} \leq 0.15 \quad (15)$$

where  $Q_{avg}$  is the average of  $Q_{hot}$  and  $Q_{cold}$ .

## 6. Data Analysis

It is requested that you include the following information in a technical report:

- For each run, use the inlet and outlet temperature readings and the flowrate readings to determine the respective energy gain/loss of the annular flow and the tube flows. Perform



an energy balance on the hot/cold streams for all trials. Does the amount of heat transferred from the hot stream equal the amount of heat absorbed by the cold stream? Should it? Explain.

- Plot the temperature profile for each tested heat transfer configuration and discuss.
- Calculate the overall heat transfer coefficient and demonstrate its dependence on the flowrate by plotting  $1/UA$  vs.  $1/v^{0.8}$  for both sets of hot and cold flow data. The plots should display  $1/UA$  vs.  $1/v^{0.8}$  for the same runs as were plotted in 1).
- Perform the Wilson analysis and determine important values (e.g.  $\alpha_i$ ,  $h_o$ ) and their respective ranges of uncertainties. Discuss the findings and compare the results with the corresponding values calculated from physical properties (evaluate at the average of the inlet and outlet temperatures).
- Calculate the value of heat transfer resistances and their uncertainties. Which term(s) in Equation 5 caused the most resistance to heat transfer?
- Determine which variables (if any) change between co-current and counter-current trials and to what degree. Which mode is most efficient?

## References

1. "Perry's Chemical Engineers' Handbook," by R. H. Perry and D. Green, Sixth Edition, pp 10-14 to 10-20, McGraw-Hill: New York (1984).
2. H. Bengtson and L. Stonecypher (Ed.), "Double Pipe Heat Exchanger Design," located at <http://www.brighthub.com/engineering/mechanical/articles/64548.aspx>, last accessed May 17, 2012.
3. J. Brown, A. Busaileh, D. French, D. Garrison and J. Johnson, "Heat Exchanger Experiment," CHE 330 Student Report (Spring 2012).
4. "Boundary layer flow" in McGraw-Hill Encyclopedia of Science and Technology, 5th edition, McGraw-Hill: New York (2004).

## Nomenclature

| Notation              | Definition                                                                                          |
|-----------------------|-----------------------------------------------------------------------------------------------------|
| $A$                   | Heat exchange surface area upon which the overall heat transfer coefficient $U$ is based            |
| $A_i$                 | Heat exchange area defined by the inner surface area of the inner tube                              |
| $A_o$                 | Heat exchange area defined by the outer surface area of the inner tube                              |
| $A_w$                 | Mean heat exchange area of the inner tube = $(A_i + A_o)/2$                                         |
| $\alpha$              | Proportionality constant defined in Equation 10                                                     |
| $\alpha_i$            | Proportionality constant for the tube flow, see Equation 11                                         |
| $\alpha_o$            | Proportionality constant for the annular flow, see Equation 12                                      |
| $C_i$                 | Intercept on Wilson plot for tube flow, see Equation 11                                             |
| $C_o$                 | Intercept on Wilson plot for annular flow, see Equation 12                                          |
| $\hat{C}_p$           | Specific heat of the hot stream or generalized specific heat in Equation 7                          |
| $\hat{C}_p'$          | Specific heat of the cold stream                                                                    |
| $h$                   | Local convective heat-transfer coefficient defined in Equation 7                                    |
| $h_i$                 | Local convective heat-transfer coefficient on the inner (or hot) side of the heat exchange surface  |
| $h_o$                 | Local convective heat-transfer coefficient on the outer (or cold) side of the heat exchange surface |
| $k$                   | Thermal conductivity                                                                                |
| $k_w$                 | Thermal conductivity of the heat exchange tube material                                             |
| $\dot{m}$             | Mass flowrate of the hot stream                                                                     |
| $\dot{m}'$            | Mass flowrate of the cold stream                                                                    |
| $\dot{Q}$             | Heat transfer rate                                                                                  |
| $R_i$                 | Heat transfer resistance due to surface scale on the inner heat exchange surface                    |
| $R_o$                 | Heat transfer resistance due to surface scale on the outer heat exchange surface                    |
| $t_w$                 | Thickness of the heat exchange wall                                                                 |
| $T_o$                 | Inlet temperature of the hot stream                                                                 |
| $T_o'$                | Inlet temperature of the cold stream                                                                |
| $T_e$                 | Outlet temperature of the hot stream                                                                |
| $T_e'$                | Outlet temperature of the cold stream                                                               |
| $U$                   | Overall heat transfer coefficient                                                                   |
| $\langle v \rangle$   | Average flow velocity                                                                               |
| $\langle v_i \rangle$ | Average flow velocity in the inner tube                                                             |
| $\langle v_o \rangle$ | Average flow velocity in the annular space                                                          |
| $\Delta T_{ln}$       | Log mean temperature difference defined by Equation (2), (6) or (8)                                 |
| $\rho$                | Fluid density                                                                                       |
| $\mu$                 | Fluid viscosity                                                                                     |